A MINI PROJECT REPORT ON

SMART STUDY PLANNER

A web-based application for organizing study schedules, tracking tasks, and improving academic productivity

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING / TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the mini project report entitled “SMART STUDY PLANNER” is a Bonafide record of work done by KARTHIK B (Register No: 922525104070) in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering/Technology in Computer Science and Engineering.

The project report has been approved as it satisfies the academic requirements with respect to the mini project work prescribed for the said degree.

Guide Name   Head of Department

MR. DR. P. ANBUMANI

Submitted for the Viva-Voce examination held on ____________________

Internal Examiner
External Examiner

DECLARATION

I hereby declare that the mini project report entitled "SMART STUDY PLANNER" submitted for the degree of Bachelor of Engineering/Technology is my original work and that it has not formed the basis for the award of any degree, diploma, associateship, fellowship, or similar title of recognition.

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KARTHIK B

ABSTRACT

In today's fast-paced academic environment, students often struggle to manage their study schedules, assignments, and examinations effectively due to poor time management and lack of organized planning tools. The “Smart Study Planner” is a web-based application designed to help students plan, track, and manage their academic tasks efficiently in a single, easy-to-use interface.

The system allows users to create study schedules, set task priorities and deadlines, track progress through a visual dashboard, and receive reminders for upcoming tasks and exams. It is built using HTML, CSS, and JavaScript on the front end, with browser-based local storage used to persist user data without requiring a dedicated back-end server, making it lightweight and easy to deploy.

Key features include a task/subject manager, a calendar-based schedule view, a to-do list with priority tagging, progress tracking with completion statistics, and a Pomodoro-style study timer to encourage focused study sessions. The application aims to reduce procrastination, improve time management skills, and promote consistent study habits among students.

The project was developed following a structured Software Development Life Cycle (SDLC) approach involving requirement analysis, design, implementation, and testing. The resulting system is responsive, user-friendly, and can be extended in the future with features such as cloud synchronization, collaborative study groups, and AI-based personalized study recommendations.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Effective time management is one of the most critical skills for academic success, yet many students continue to rely on scattered notes, sticky reminders, or memory alone to keep track of assignments, exams, and study sessions. The Smart Study Planner is a web-based application built to bring structure to this process by offering a single platform where students can plan subjects, tasks, and schedules, monitor their progress, and stay on top of deadlines.

1.2 Problem Statement

Students frequently face difficulty balancing multiple subjects, assignment deadlines, and examination preparation simultaneously. The lack of a centralized, easy-to-use planning tool often leads to missed deadlines, last-minute cramming, and increased academic stress. There is a need for a lightweight, accessible application that helps students organize their study routine, prioritize tasks, and track their progress without requiring complex installation or paid subscriptions.

1.3 Objectives

- To design a simple and intuitive interface for creating and managing study tasks and schedules.
- To allow students to set priorities and deadlines for assignments and exams.
- To provide a visual dashboard that displays progress and pending tasks at a glance.
- To include a focused study timer (Pomodoro technique) to improve concentration.
- To store data locally in the browser so the application works without a dedicated server or internet dependency.

1.4 Scope of the Project

The Smart Study Planner is intended for individual students at the school or college level who wish to organize their personal study routine. The current scope covers task creation and management, a calendar/schedule view, progress tracking, and a study timer. It does not, in its present form, provide multi-user collaboration, cloud synchronization, or mobile application support — these are identified as directions for future enhancement.

1.5 Organization of the Report

This report is organized into eight chapters. Chapter 1 introduces the project. Chapter 2 presents a survey of existing study-planning tools and related literature. Chapter 3 discusses the system analysis, including the existing system, proposed system, and feasibility study. Chapter 4 describes the system design, including architecture and data flow. Chapter 5 explains the implementation details and technologies used. Chapter 6 covers the testing strategy and test cases. Chapter 7 presents the results and screenshots of the working application. Chapter 8 concludes the report and outlines scope for future enhancement.

CHAPTER 2

LITERATURE SURVEY

2.1 Existing Study Planning Tools

A number of digital tools currently exist to help individuals manage tasks and schedules, including general-purpose to-do list applications, calendar applications, and dedicated study-planner mobile apps. General productivity tools such as calendar and reminder applications are widely used but are not tailored specifically to the academic needs of students, such as subject-wise organization or exam-focused planning. Dedicated study-planner mobile applications provide more relevant features but often require account creation, internet connectivity, or paid subscriptions, which can be a barrier for many students.

2.2 Research Observations

A review of existing academic productivity tools and related student surveys reveals a few consistent patterns. Students prefer simple, visually clear interfaces over feature-heavy applications that are difficult to navigate. Progress-tracking indicators, such as completion percentages, have been found to be positively associated with sustained motivation. In addition, time-boxed study methods such as the Pomodoro technique are widely recommended in academic productivity literature for improving focus and reducing burnout during long study sessions.

2.3 Summary

The literature survey indicates a clear opportunity for a lightweight, browser-based study planner that combines task management, scheduling, and a focus timer within a single, distraction-free interface. This observation directly informed the design and feature set of the Smart Study Planner, which is described in the following chapters.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

Currently, most students rely on manual methods such as physical planners, sticky notes, or generic mobile reminder applications to manage their academic responsibilities. These methods have several limitations: they are not centralized, provide no visual representation of overall progress, offer no structured way to prioritize tasks by urgency or importance, and do not support focused study techniques. As a result, students often experience disorganized preparation and increased stress closer to deadlines.

3.2 Proposed System

The proposed Smart Study Planner addresses these limitations by providing a single web-based platform where students can add subjects and tasks, assign deadlines and priority levels, view their schedule in a calendar format, and track overall progress through a visual dashboard. The system also includes a built-in Pomodoro-style timer to support focused study sessions. All data is stored locally within the browser, so the application requires no server setup, account registration, or internet connection once loaded, making it fast, private, and easy to use.

3.3 Feasibility Study

3.3.1 Technical Feasibility

The system is built using standard web technologies — HTML, CSS, and JavaScript — which are well-documented, widely supported by all modern browsers, and require no specialized hardware or software licenses. This makes the project technically feasible to develop, test, and deploy.

3.3.2 Economic Feasibility

Since the application relies entirely on open, free web technologies and browser-based local storage, there are no licensing costs, server hosting fees, or database costs involved, making the project highly economical to build and maintain.

3.3.3 Operational Feasibility

The application has a simple, intuitive interface that requires no special training to operate. Any student with basic computer literacy and access to a web browser can use the Smart Study Planner effectively.

3.4 Software Requirements

Component	Specification
Operating System	Windows 10/11, macOS, or Linux (any OS with a modern browser)
Front-End Technologies	HTML5, CSS3, JavaScript (ES6)
Browser	Google Chrome / Mozilla Firefox / Microsoft Edge (latest version)
Code Editor	Visual Studio Code
Data Storage	Browser LocalStorage API
Version Control (optional)	Git and GitHub

3.5 Hardware Requirements

Component	Minimum Requirement
Processor	Dual-core 1.6 GHz or higher
RAM	2 GB or more
Storage	100 MB free disk space
Display	1024 x 768 resolution or higher
Internet	Required only for initial download; not required afterwards

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The Smart Study Planner follows a simple client-side architecture. The entire application runs within the user's web browser, with three main layers: the Presentation Layer (HTML/CSS for structure and styling), the Application Logic Layer (JavaScript for handling user interactions, task management, and timer logic), and the Data Storage Layer (the browser's LocalStorage API, which persists tasks, schedules, and settings between sessions).

Architecture Diagram (described): User Interface (Browser) → JavaScript Application Logic (Task Manager, Scheduler, Timer, Dashboard) → LocalStorage (Data Persistence). All operations happen on the client side; no external server or database is involved.

4.2 Use Case Diagram

The primary actor in the system is the student, who interacts with the application to perform the following use cases: Add Subject/Task, Set Priority and Deadline, View Calendar/Schedule, Mark Task as Complete, View Progress Dashboard, Start/Stop Study Timer, Delete/Edit Task, and Receive Reminders for upcoming deadlines.

4.3 Data Flow Diagram

4.3.1 Level 0 DFD (Context Diagram)

Student → [Smart Study Planner System] → Task List, Schedule View, Progress Report. The student provides task and schedule inputs to the system, and the system returns organized views, reminders, and progress statistics.

4.3.2 Level 1 DFD

- Process 1: Task Input — captures subject, task name, priority, and deadline; stores it via the Data Storage process.
- Process 2: Schedule Management — organizes tasks by date and displays them in a calendar view.
- Process 3: Progress Tracking — calculates completed vs. pending tasks and updates the dashboard.
- Process 4: Timer Module — manages Pomodoro study/break intervals independently of the task data.

4.4 Database Design

As the application uses browser LocalStorage rather than a relational database, data is stored in structured JSON format. The primary data entity is the Task object, structured as follows:

Field Name	Data Type	Description
Task Id	String	Unique identifier for the task
subject	String	Subject or course name
title	String	Task title or description
priority	String	High / Medium / Low
Due Date	Date	Deadline for the task
status	String	Pending / In Progress / Completed
Created At	Date	Timestamp when the task was created

4.5 Module Description

4.5.1 Task Management Module

Allows the student to add, edit, delete, and mark tasks as complete, with fields for subject, priority, and due date.

4.5.2 Schedule/Calendar Module

Displays all tasks in a calendar format, allowing the student to view tasks due on a specific date at a glance.

4.5.3 Dashboard Module

Presents summary statistics such as total tasks, completed tasks, pending tasks, and completion percentage using simple visual indicators.

4.5.4 Study Timer Module

Implements a Pomodoro-style timer with configurable focus and break intervals to encourage concentrated study sessions.

CHAPTER 5

IMPLEMENTATION

5.1 Technologies Used

- HTML5 — for structuring the web pages and content.
- CSS3 — for styling, layout, and responsive design.
- JavaScript (ES6) — for application logic, DOM manipulation, and event handling.
- LocalStorage API — for persisting task and schedule data in the browser.
- Visual Studio Code — as the development environment.

5.2 Implementation Details

The application is implemented as a single-page interface divided into sections: a task-entry form, a task list view, a calendar/schedule view, a progress dashboard, and a study timer. When a student adds a new task, the JavaScript application validates the input, generates a unique task ID, and stores the task object as a JSON string in LocalStorage. On every page load, the application reads existing data from LocalStorage and dynamically renders the task list, calendar, and dashboard statistics using DOM manipulation.

Task priority is visually indicated using color coding (for example, red for high priority, yellow for medium, and green for low), helping students quickly identify urgent work. The dashboard recalculates completion percentage in real time whenever a task's status changes.

5.3 Sample Code

The following snippet illustrates how a new task is added and saved to LocalStorage:

```
function addTask(subject, title, priority, dueDate) {   const tasks =
JSON.parse(localStorage.getItem('tasks')) || [];   const newTask = {
taskId: Date.now().toString(),     subject, title, priority, dueDate,
status: 'Pending',     createdAt: new Date().toISOString()   };
tasks.push(newTask);   localStorage.setItem('tasks',
JSON.stringify(tasks));   renderTasks();   updateDashboard(); }
```

The following snippet illustrates the Pomodoro timer logic used in the Study Timer Module:

```
let timeLeft = 25 * 60; let timerId = null;  function startTimer() {
timerId = setInterval(() => {     if (timeLeft <= 0) {
clearInterval(timerId);       alert('Focus session complete! Take a
break.');       return;     }     timeLeft--;
updateTimerDisplay(timeLeft);   }, 1000); }
```

CHAPTER 6

TESTING

6.1 Testing Objectives

The primary objective of testing was to ensure that the Smart Study Planner functions correctly across common user interactions, handles invalid input gracefully, correctly persists data using LocalStorage, and displays accurate progress statistics.

6.2 Types of Testing

6.2.1 Unit Testing

Individual JavaScript functions, such as addTask(), deleteTask(), and updateDashboard(), were tested independently to verify that each performs its intended operation correctly.

6.2.2 Integration Testing

The interaction between the task management module, the calendar view, and the dashboard module was tested to ensure that changes in one module (e.g., marking a task complete) are correctly reflected across the others.

6.2.3 System Testing

The complete application was tested as a whole to confirm that all modules work together seamlessly from task creation to progress tracking.

6.2.4 User Acceptance Testing

A small group of students used the application and provided feedback on usability, interface clarity, and overall satisfaction, which was used to refine the final design.

6.3 Test Cases

Test ID	Test Description	Input	Expected Output	Result
TC01	Add new task with valid data	Subject, title, priority, due date	Task added and displayed in task list	Pass
TC02	Add task with empty title field	Empty title	Validation error shown; task not added	Pass
TC03	Mark task as completed	Click 'Complete' on a task	Task status updates; dashboard refreshes	Pass
TC04	Delete an existing task	Click 'Delete' on a task	Task removed from list and storage	Pass

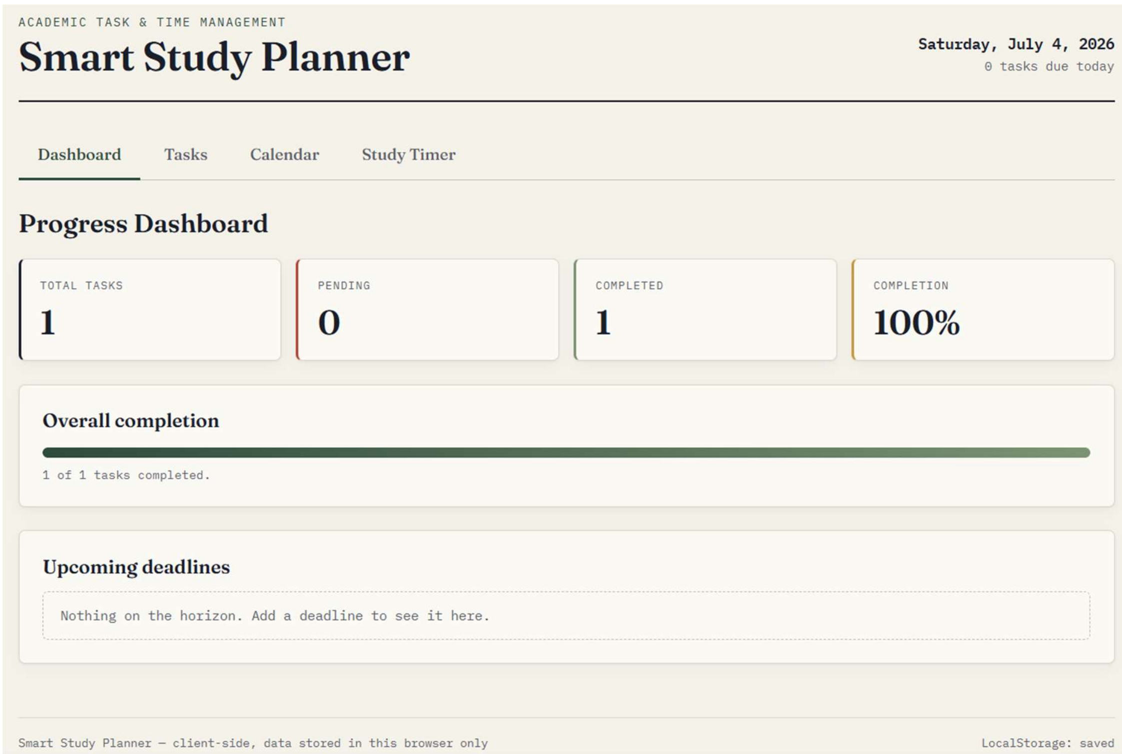
Test ID	Test Description	Input	Expected Output	Result
TC05	Reload page after adding tasks	Browser refresh	Previously added tasks are still displayed	Pass
TC06	Start Pomodoro timer	Click 'Start Timer'	Countdown begins from set duration	Pass
TC07	View dashboard after completing tasks	Multiple tasks marked complete	Completion percentage updates correctly	Pass

CHAPTER 7

RESULTS AND SCREENSHOTS

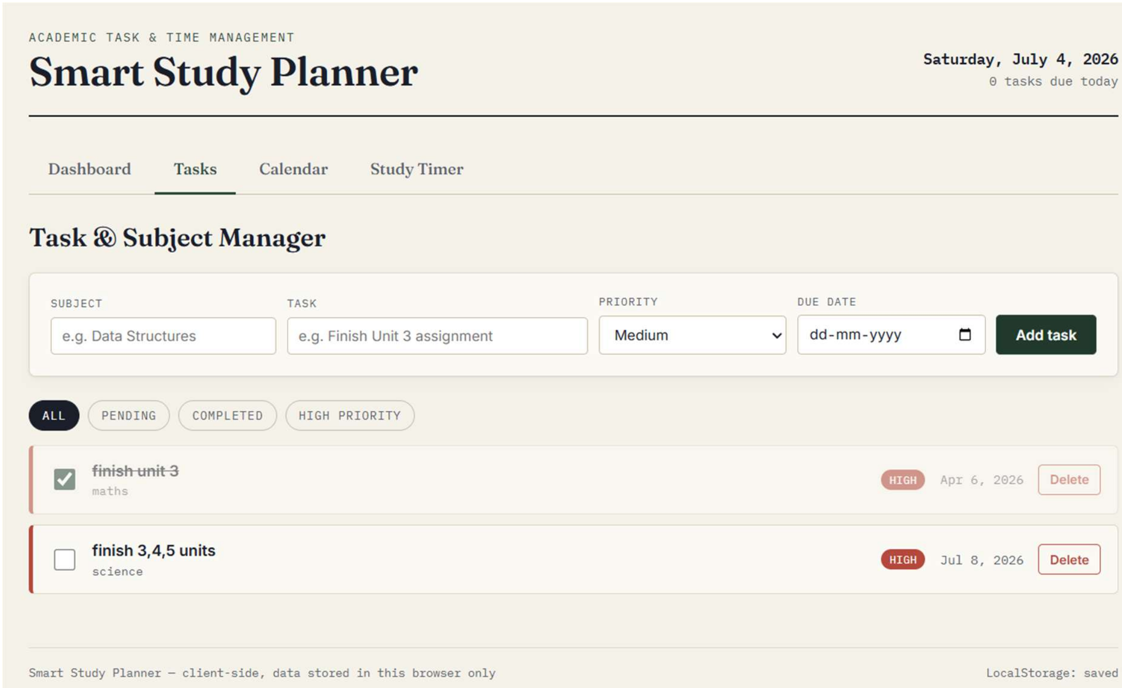
This chapter presents the working interface of the Smart Study Planner application. [Insert screenshots of your running application below each caption after you build/run the project.]

7.1 Home / Task Dashboard



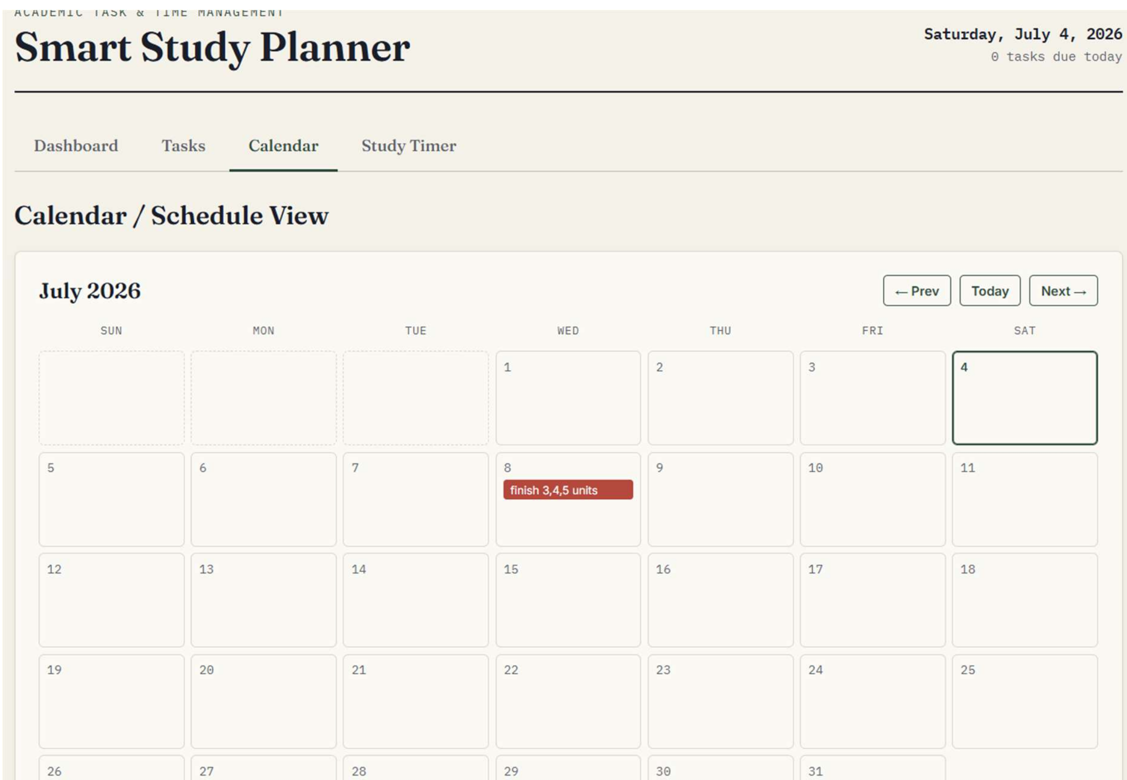
Displays the overall task list, priority indicators, and progress summary.

7.2 Add Task Form



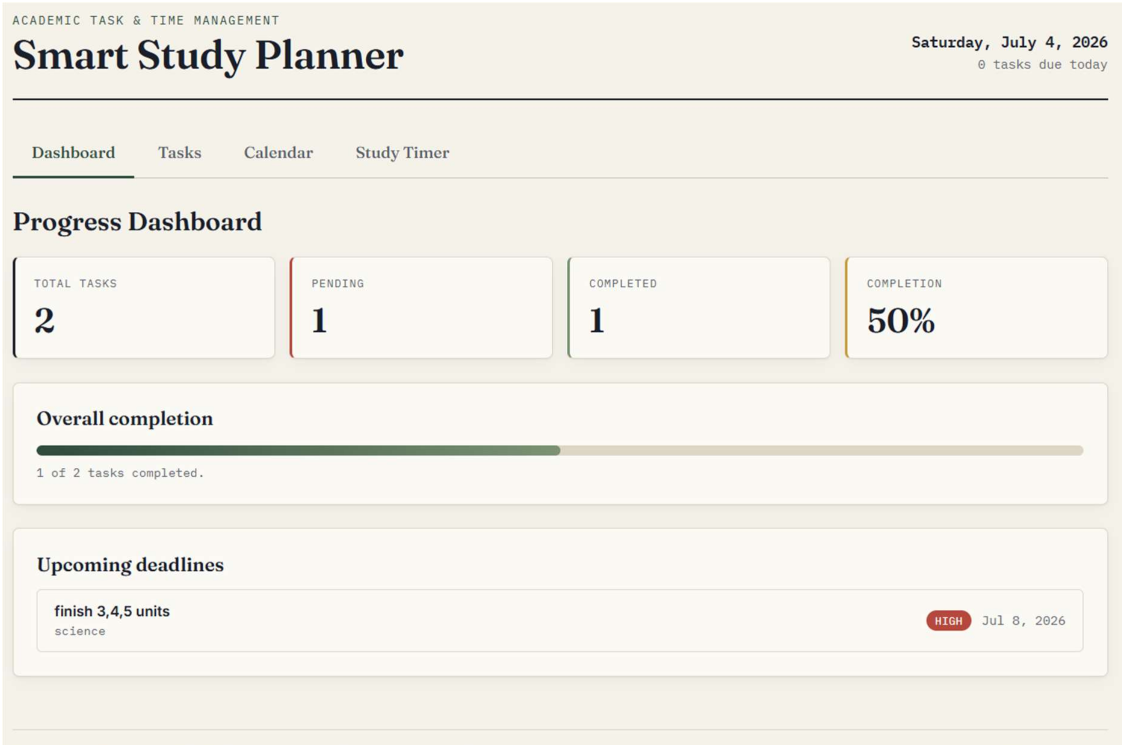
Shows the form used to add a new subject/task with priority and deadline.

7.3 Calendar / Schedule View



Displays tasks organized by date in a calendar layout.

7.4 Progress Dashboard



Shows completion statistics such as total, pending, and completed tasks.

7.5 Study Timer

Displays the Pomodoro-style focus timer interface.

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENT

8.1 Conclusion

The Smart Study Planner successfully provides students with a simple, lightweight, and effective tool for organizing their academic tasks and study schedules. By combining task management, calendar scheduling, progress tracking, and a focused study timer within a single browser-based interface, the application addresses common challenges students face in managing their time and staying consistent with their studies. The project was developed using standard web technologies, making it accessible, cost-free, and easy to maintain, while fulfilling all the objectives outlined at the start of the project.

8.2 Future Enhancements

- Cloud synchronization and user accounts to allow access across multiple devices.
- Collaborative features enabling group study planning and shared task lists.
- AI-based personalized study recommendations based on task history and performance.
- Push notifications and email/SMS reminders for upcoming deadlines.
- A dedicated mobile application version using a framework such as React Native or Flutter.
- Integration with academic calendars and learning management systems (LMS).

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